Unit 4 Topic Questions

1. An object of mass $m = 0.5\ \text{kg}$ is moving in the xy-plane with velocity components of $v_x = 3.0\ \text{m/s}$ and $v_y = -4.0\ \text{m/s}$. What is the magnitude of the object's momentum?

(A) $0.5\ \text{kg·m/s}$

(B) $2.5\ \text{kg·m/s}$

(C) $3.5\ \text{kg·m/s}$

(D) $5.0\ \text{kg·m/s}$

2. A moving block of mass m_0 has kinetic energy K_0 and momentum of magnitude p_0. A second block of mass $2m_0$ is moving with the same kinetic energy K_0. What is the magnitude of the momentum of the second block?

(A) $2p_0$

(B) $\frac{2p_0}{\sqrt{2}}$

(C) p_0

(D) $\frac{p_0}{\sqrt{2}}$

3. The kinetic energy of a moving object changes from K_0 to $4K_0$. By what factor does the magnitude of the object's momentum change?

(A) 1

(B) 2

(C) 4

(D) 16

4. A block with a mass of $2.0\ \text{kg}$ starts from rest and begins to move along the x-axis at time $t = 0$ according to the equation $x = \frac{1}{2}t^3$, where x is the block's position, in meters, and t is the time, in seconds. What is the momentum of the block at time $t = 4.0\ \text{s}$?

(A) $16\ \text{kg} \cdot \text{m/s}$

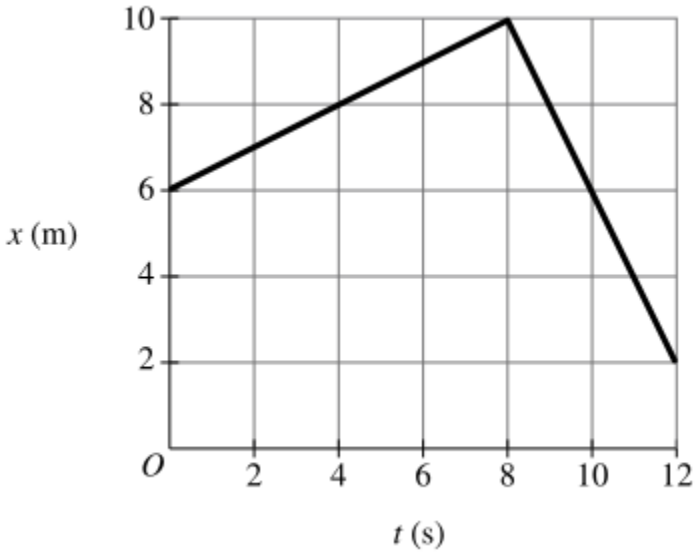
(B) $24\ \text{kg} \cdot \text{m/s}$

(C) $48\ \text{kg} \cdot \text{m/s}$

(D) $64\ \text{kg} \cdot \text{m/s}$

Unit 4 Topic Questions

5.



The graph shows the position x as a function of time t for a cart of mass $m = 3.0\ \text{kg}$ that moves along the x-axis. What is the magnitude of the change in the cart's momentum between $t = 4\ \text{s}$ and $t = 10\ \text{s}$?

(A) $1.0\ \text{kg·m/s}$

(B) $2.5\ \text{kg·m/s}$

(C) $4.5\ \text{kg·m/s}$

(D) $7.5\ \text{kg·m/s}$

6. A block slides along a surface with a speed v_0 and a momentum of magnitude p_0. It collides with and sticks to an identical block that is initially at rest. Immediately after the collision, the two blocks slide together with a speed of $\frac{1}{2}v_0$. What is the magnitude of the momentum of the two-block system immediately after the collision?

(A) $\frac{1}{4}p_0$

(B) $\frac{1}{2}p_0$

(C) p_0

(D) $2p_0$

7. Two moving objects each have momenta of equal magnitude p_0. Object 1 is more massive than Object 2. Which of the following claims about the speed of Object 2 is true?

(A) It is greater than the speed of Object 1.

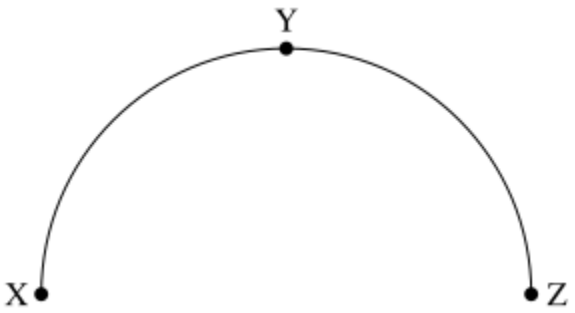
(B) It is equal to the speed of Object 1.

(C) It is less than the speed of Object 1.

(D) It could be greater than, equal to, or less than the speed of Object 1 depending on the directions of the momenta.

Unit 4 Topic Questions

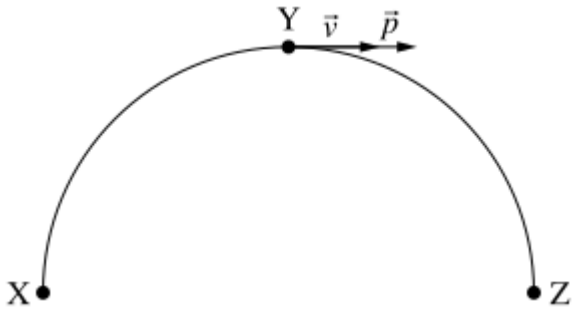
8.



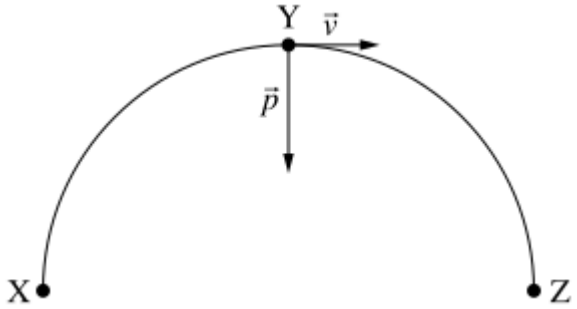
A semicircular path connects three points labeled X, Y, and Z, as shown in the figure. A particle is initially at point X and moves along the path with a constant speed toward point Z. When the particle is at point Y, as shown in the figure, which of the following diagrams correctly represents the directions of the velocity $\overrightarrow{v}$ and momentum $\overrightarrow{p}$ of the object?

Unit 4 Topic Questions

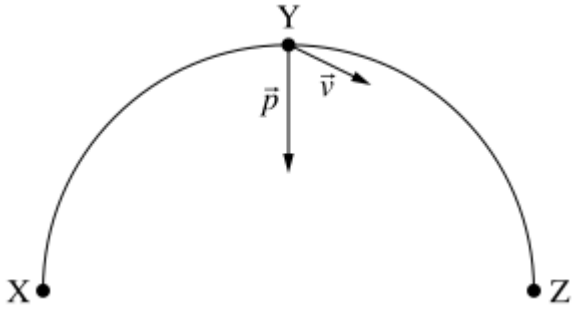
(A)



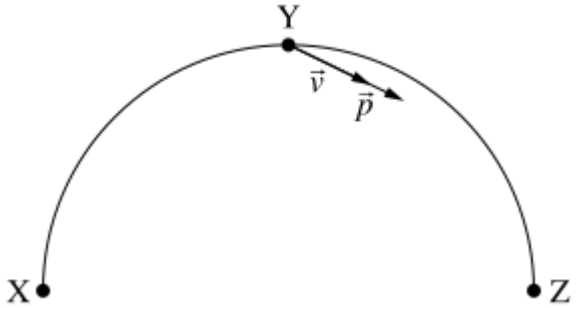
(B)



(C)



(D)



9. A student is pulling a cart full of water horizontally on flat ground at a constant velocity. The cart has a small hole through which the water slowly leaks out of the cart. For the system consisting of the cart and the water within the cart, how is the magnitude of the momentum of the system changing, if at all?

(A) The magnitude of the momentum is increasing.

(B) The magnitude of the momentum remains the same.

(C) The magnitude of the momentum is decreasing.

(D) The answer cannot be determined without knowing the cart's mass and the mass of water that leaks through the hole each second.

Unit 4 Topic Questions

10. A 50-kg student sitting in a rolling chair at rest pushes against a wall, which applies a $10\ \text{N}\cdot\text{s}$ horizontal impulse to the student. Later, a 40-kg student is at rest in the same rolling chair and catches a 10-kg ball while applying a $10\ \text{N}\cdot\text{s}$ horizontal impulse to the ball. Which of the following statements best describes the students' final speeds and provides valid reasoning supporting the claim?

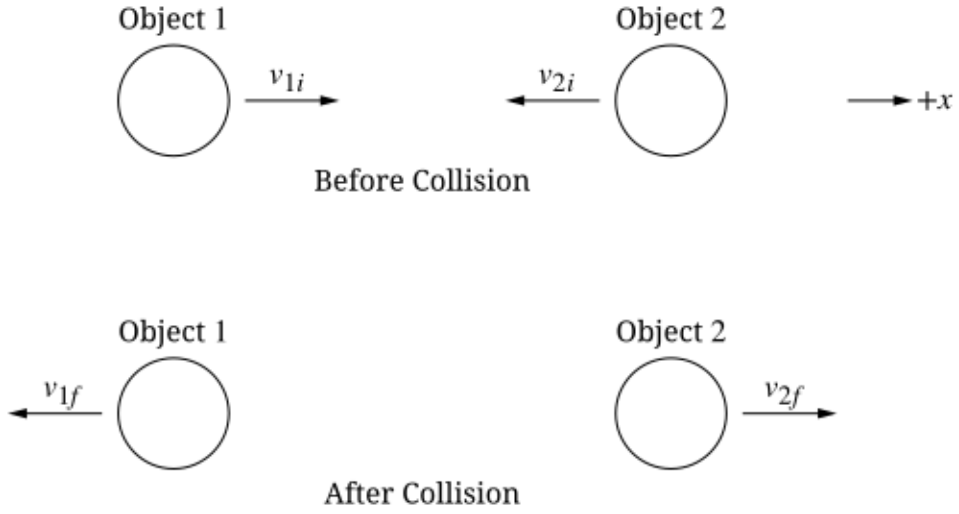
(A) Both students have the same final speed because the $50\ \text{kg}$ student-chair system and the $10\ \text{kg}$ ball-$40\ \text{kg}$ student-chair system have the same mass and experience impulses of the same magnitude.

(B) Both students have the same final speed because each ball-chair-student system experiences the same impulse regardless of the mass of the system.

(C) The 40-kg student has a greater final speed because the 40-kg student-chair system has less mass than the 50-kg student-chair system.

(D) The 40-kg student has a greater final speed because the ball transfers all of its momentum to the student when it is caught.

11.



Object 1 of mass m_1 and Object 2 of mass m_2 move along the x-axis toward each other with speeds v_{1i} and v_{2i}, respectively. The objects collide and then move away from each other with speeds v_{1f} and v_{2f}, respectively, as shown in the figure. There are no external forces acting on the two-object system. What is the magnitude of the impulse exerted on Object 1 by Object 2 during the collision?

(A) $m_1|v_{1f} - v_{1i}|$

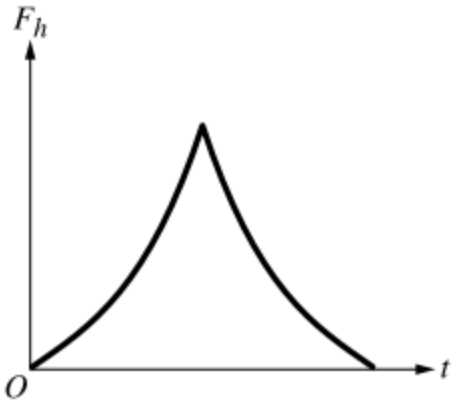
(B) $m_1(v_{1f} + v_{1i})$

(C) $m_1|v_{2f} - v_{2i}|$

(D) $m_1(v_{2f} + v_{2i})$

Unit 4 Topic Questions

12.



A piece of clay is thrown against a hard wall. The graph shows the horizontal component F_h of the net force exerted on the clay during the collision as a function of time t. How would the maximum of F_h, the duration of the collision, and the area under the graph change if an identical piece of clay was instead thrown against a padded wall with the same velocity? Assume that both collisions are perfectly inelastic.

	Maximum of F_h	**Duration of collision**	**Area under graph**
A	Increases	Decreases	Increases
B	Decreases	Increases	Remains the same
C	Increases	Decreases	Remains the same
D	Decreases	Remains the same	Decreases

(A) A

(B) B

(C) C

(D) D

13. A force F that varies with time t is exerted on an object. The force is described by the equation $F(t) = F_0 e^{-bt}$ where F_0 and b are positive constants. If the object is at rest at time $t = 0$, the momentum of the object a very long time later is most nearly

(A) zero

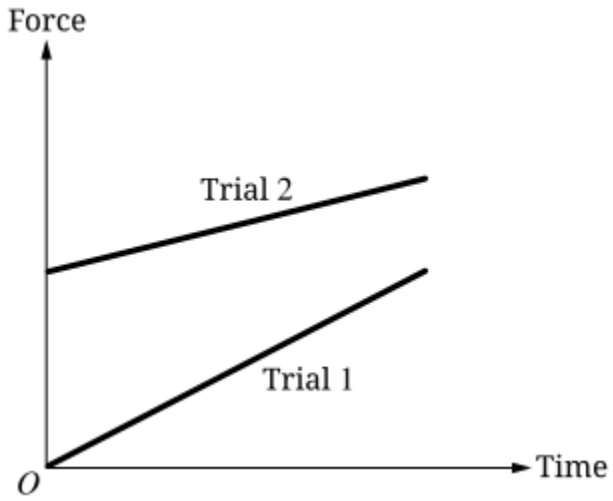
(B) bF_0

(C) $\frac{F_0}{b}$

(D) $-\frac{F_0}{b}$

Unit 4 Topic Questions

14.



Students are conducting an experiment on a block that is free to move along a straight, horizontal track with negligible friction. They conduct two trials, Trial 1 and Trial 2, in which the block is initially at rest and a time-varying horizontal force is applied to the block in a direction parallel to the track. The applied force as a function of time is shown in the graph for each trial, both of which last the same amount of time. During Trial 1, the students do work W_1 on the block and apply an impulse J_1. During Trial 2, they do work W_2 on the block and apply an impulse J_2. How do these values of work and impulse compare to each other?

	Work	Impulse
A	$W_1 < W_2$	$J_1 < J_2$
B	$W_1 < W_2$	$J_1 > J_2$
C	$W_1 > W_2$	$J_1 < J_2$
D	$W_1 > W_2$	$J_1 > J_2$

(A) A

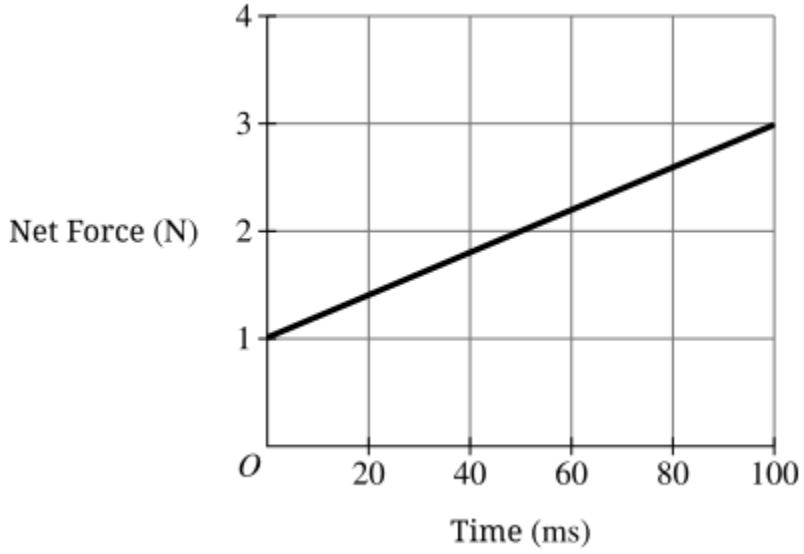
(B) B

(C) C

(D) D

Unit 4 Topic Questions

15.



A lab cart is traveling along a straight, horizontal track with an unknown initial velocity. The graph shows the net force on the cart as a function of time during a short time interval. Does the graph provide sufficient information to determine the change in momentum of the cart during the time interval shown, and what evidence supports the claim?

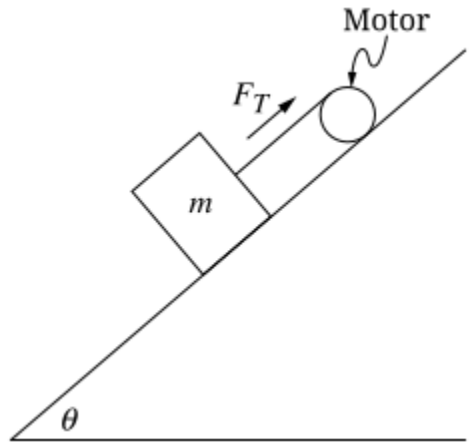
(A) Yes, because the change in momentum is equal to the area under the curve of the force-versus-time graph.

(B) Yes, because the change in momentum is equal to the slope of the force-versus-time graph.

(C) No, because the initial momentum is unknown and therefore the change in momentum cannot be determined.

(D) No, because the force is not constant and therefore the change in momentum cannot be determined.

16.



A block of mass m can move with negligible friction on a ramp that is at an angle θ from the horizontal, as shown in the figure. Starting at time $t = 0$, the motor creates a varying string tension F_T given by $F_T = 2kt$, where k is a positive constant. If the positive direction is taken to be up the ramp, what is the magnitude of the impulse exerted on the block between time $t_1 > 0$ and a later time t_2 ?

Unit 4 Topic Questions

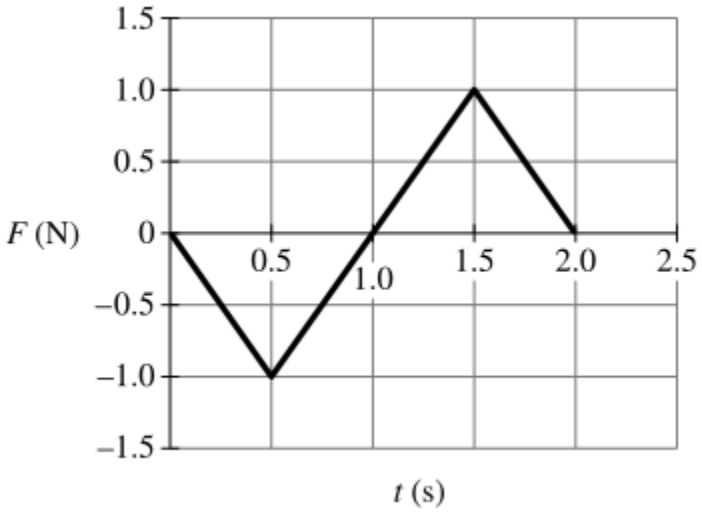
(A) $k\left({t_2}^2 - {t_1}^2\right)$

(B) $k{t_2}^2 - mg\sin\theta(t_2)$

(C) $k\left({t_2}^2 - {t_1}^2\right) - mg\sin\theta(t_2 - t_1)$

(D) $k\left({t_2}^2 - {t_1}^2\right) - mg(t_2 - t_1)$

17.



The graph shows the net force F exerted on an object as a function of time t. How does the impulse J_1 exerted on the object from $t = 0$ s to $t = 1$ s compare to the impulse J_2 exerted on the object from $t = 1$ s to $t = 2$ s ?

(A) $J_1 = -J_2$

(B) $J_1 = J_2$

(C) $|J_1| < |J_2|$

(D) $|J_1| > |J_2|$

18. A toy water launcher, initially full of water, can slide with negligible friction on a horizontal surface. Water is ejected horizontally from the launcher with a constant speed relative to the launcher. The mass of water ejected from the launcher over any time interval of length Δt is constant as well. Which of the following statements correctly describes how the launcher moves, if at all, and provides a correct justification?

(A) The launcher will remain in place because there is no horizontal force acting on the launcher.

(B) The launcher will move at a constant speed because the water is ejected at a constant speed.

(C) The launcher will move at a constant acceleration because the system of the launcher and remaining water experiences the same impulse during any time interval of length Δt.

(D) The launcher will move at an increasing acceleration because the mass of the system of the launcher and remaining water is decreasing, and the system experiences the same impulse during any time interval of length Δt.

Unit 4 Topic Questions

19. In three different scenarios, a student of mass $3m$ pulls on a lightweight rope attached to a block of mass m. The student and the block are both initially at rest in each scenario. In Scenario 1, the student stands on the shore of a frozen lake and the block is on the ice. After the student starts pulling on the rope in Scenario 1, the block slides with negligible friction while the student remains in the same location on the shore. In Scenario 2, both the student and the block are on the ice. After the student starts pulling on the rope in Scenario 2, both the student and the block slide with negligible friction. In Scenario 3, the block is at rest on the shore while the student is on the ice. After the student starts pulling on the rope in Scenario 3, the student slides with negligible friction while the block remains in the same location on the shore. The final velocity of the block relative to the student is the same in each scenario. For which scenario, if any, is the final speed of the center of mass of the block-student system the greatest?

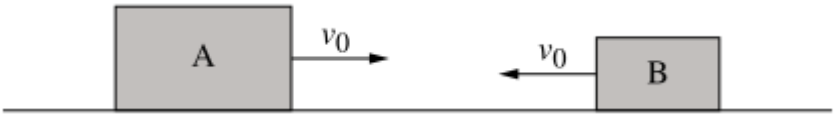
(A) Scenario 1

(B) Scenario 2

(C) Scenario 3

(D) The speed of the center of mass is the same in all three scenarios.

20.



Trial	Center of mass speed after collision	Impulse exerted on Block B
1	v_1	J_1
2	v_2	J_2

Blocks A and B can move along a horizontal surface with negligible friction, as shown in the figure. The mass of Block A is three times the mass of Block B. In two separate experimental trials, the blocks are initially moving toward each other, each with a speed of v_0. In Trial 1, Block A comes to rest after the collision, while in Trial 2 the blocks stick together. The table shows the speed of the center of mass of the two-block system after the collision in each trial, as well as the magnitude of the impulse exerted on Block B during each collision. Which of the following correctly compares these values?

	Center of mass speeds	**Magnitude of the impulses exerted on Block B**
A	$v_1 = v_2$	$J_1 > J_2$
B	$v_1 = v_2$	$J_1 = J_2$
C	$v_1 > v_2$	$J_1 > J_2$
D	$v_1 > v_2$	$J_1 = J_2$

(A) A

(B) B

(C) C

(D) D

Unit 4 Topic Questions

21.



Two blocks A and B have masses m and $2m$, respectively, and are free to move on a horizontal surface with negligible friction, as shown in the figure. Block A is initially moving with speed v_0 toward Block B, which is at rest and has a spring attached to one side. Block A then makes contact with the spring, causing the spring to compress. After the spring reaches its maximum compression, the blocks move apart and Block A loses contact with the spring. What is the speed of the center of mass of the blocks-spring system after the blocks have separated?

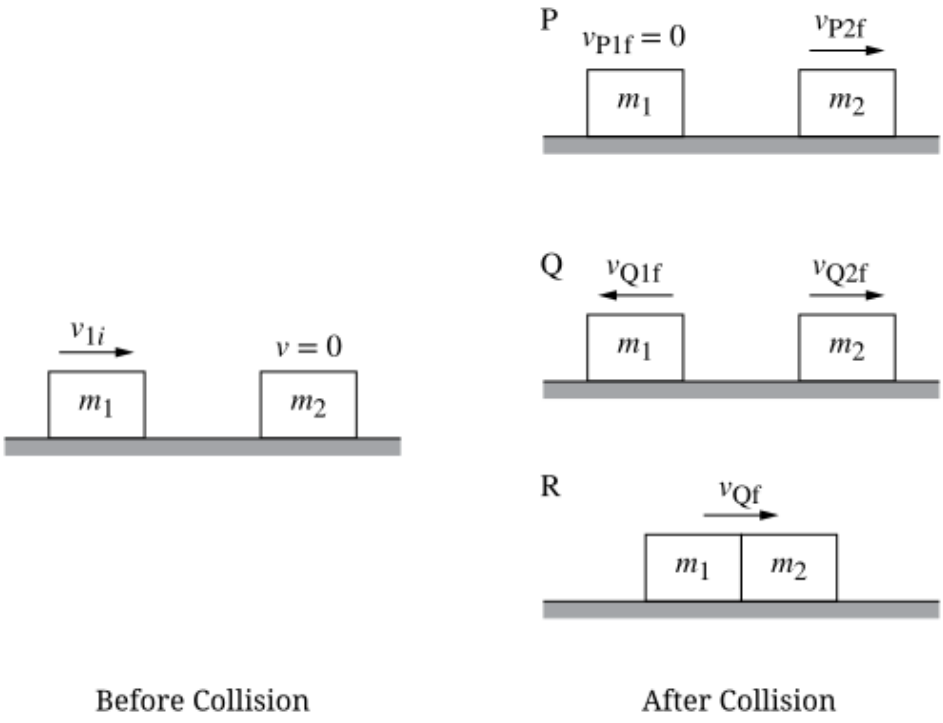
(A) v_0

(B) $\frac{2v_0}{3}$

(C) $\frac{v_0}{2}$

(D) $\frac{v_0}{3}$

22.



A block of mass m_1 moves with initial speed v_{1i} on a horizontal surface toward a stationary block of mass m_2, as shown in the figure labeled "Before Collision." Friction between the blocks and the horizontal surface is negligible. Three possible outcomes labeled P, Q, and R, are also shown. Rank the two-block system's final center of mass speeds v_{Pcm}, v_{Qcm}, and v_{Rcm} after the three collision outcomes.

Unit 4 Topic Questions

(A) $v_{\mathrm{Pcm}} = v_{\mathrm{Qcm}} = v_{\mathrm{Rcm}}$

(B) $v_{\mathrm{Qcm}} < v_{\mathrm{Rcm}} < v_{\mathrm{Pcm}}$

(C) $v_{\mathrm{Rcm}} < v_{\mathrm{Pcm}} < v_{\mathrm{Qcm}}$

(D) $v_{\mathrm{Pcm}} < v_{\mathrm{Qcm}} < v_{\mathrm{Rcm}}$

23.

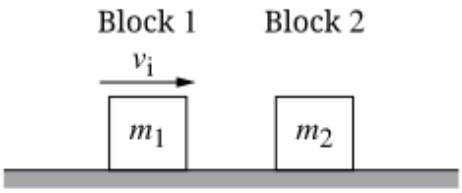


Figure 1

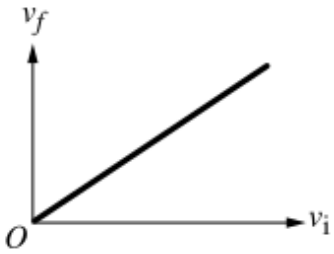


Figure 2

A collision experiment is conducted using two blocks, Block 1 and Block 2, that can slide with negligible friction along a horizontal surface. Block 1 of mass m_1 is initially moving with speed v_i toward Block 2, which has a mass m_2 and is initially at rest, as shown in Figure 1. The blocks collide, and after the collision they stick together and move with speed v_f. In multiple trials of this experiment, Block 1 is given different initial velocities. Data for v_f as a function of v_i are recorded, and the data is used to make a best-fit curve, which is a straight line as shown in Figure 2. If the experiment were repeated with Block 1 replaced by a block of mass $2m_1$, how would the best-fit line to the data change, if at all?

(A) The slope of the line would be less.

(B) The slope of the line would be greater.

(C) The line would become a curve with a slope that is not constant.

(D) The line would not change.

24. A 4-kg object moving to the left collides with and sticks to a 3-kg object moving to the right. Just before the collision, the lighter object is moving faster than the heavier object. Which of the following is true of the combined objects immediately after the collision?

(A) They must be moving to the left.

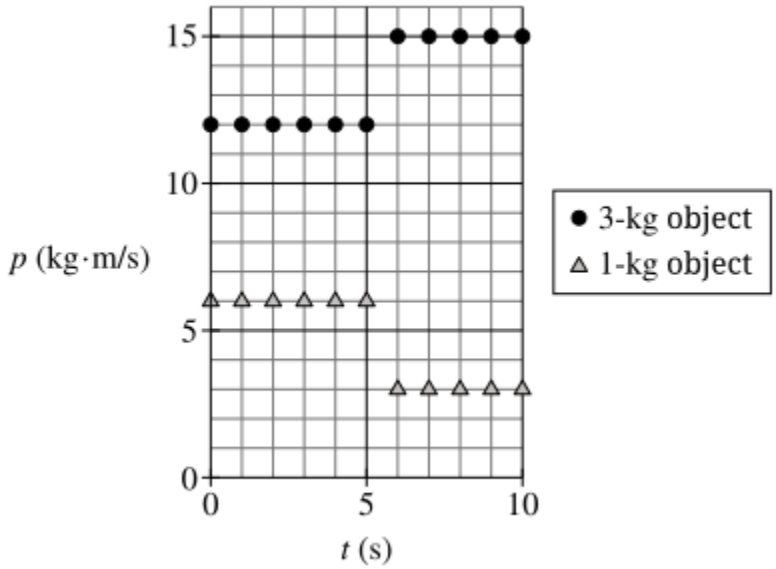
(B) They must be moving to the right.

(C) They must be at rest.

(D) The combined objects' direction of motion, if any, cannot be determined without knowing the ratio of the speeds of the objects before the collision.

Unit 4 Topic Questions

25.



A 1-kg object moving at $6\ \text{m/s}$ collides with a 3-kg object moving at $4\ \text{m/s}$ in the same direction. Data for the momentum p as a function of time t for both objects is shown in the graph. Which of the following statements best describes this collision?

(A) The momentum of the two-object system is constant, and the magnitude of the relative velocity between the objects is the same before and after the collision.

(B) The momentum of the two-object system is constant, and the magnitude of the relative velocity between the objects is different before and after the collision.

(C) The momentum of the 1-kg object is constant, and the magnitude of the relative velocity between the objects is the same before and after the collision.

(D) The momentum of the 1-kg object is not constant, and the magnitude of the relative velocity between the objects is different before and after the collision.

26. Two objects, both of mass m, are free to slide with negligible friction on a horizontal surface that lies in the plane of an xy-coordinate system. Initially one of the objects moves with a velocity v_0 in the $+x$-direction and toward the second object, which is at rest. The two objects interact for a brief time with no net external force acting on the two-object system. After the interaction the first object moves with a velocity that has positive x and y components, a magnitude v_0, and is directed at an angle θ from the $+x$-direction. What are the final x-component v_x and y-component v_y of the velocity of the object that was initially at rest?

	v_x	v_y
A	$v_0(1-\cos\theta)$	$-v_0 \sin\theta$
B	$-v_0 \cos\theta$	$-v_0 \sin\theta$
C	v_0	$-v_0$
D	0	0

Unit 4 Topic Questions

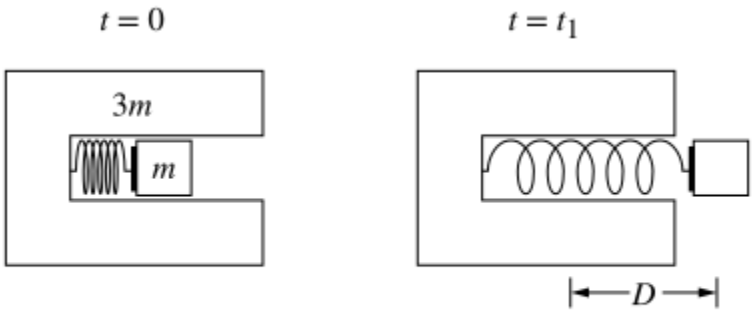
(A) A

(B) B

(C) C

(D) D

27.



Note: Figure not drawn to scale.

A small block of mass m is held in place inside a larger, U-shaped block of mass $3m$, as shown in the figure. Initially, the centers of mass of the two blocks are located at the same position, and a spring attached to the larger block is compressed a distance D from its relaxed length by the smaller block. At time $t = 0$, the two blocks are released from rest. At time $t_1 > 0$, the spring is at its relaxed length, the centers of mass of the two blocks are a distance D apart, and the small block is moving to the right with a velocity v just as it loses contact with the spring. If there are no external forces exerted on the blocks-spring system, what is the distance of separation of the two centers of mass at a later time $t > t_1$?

(A) $4v(t - t_1)$

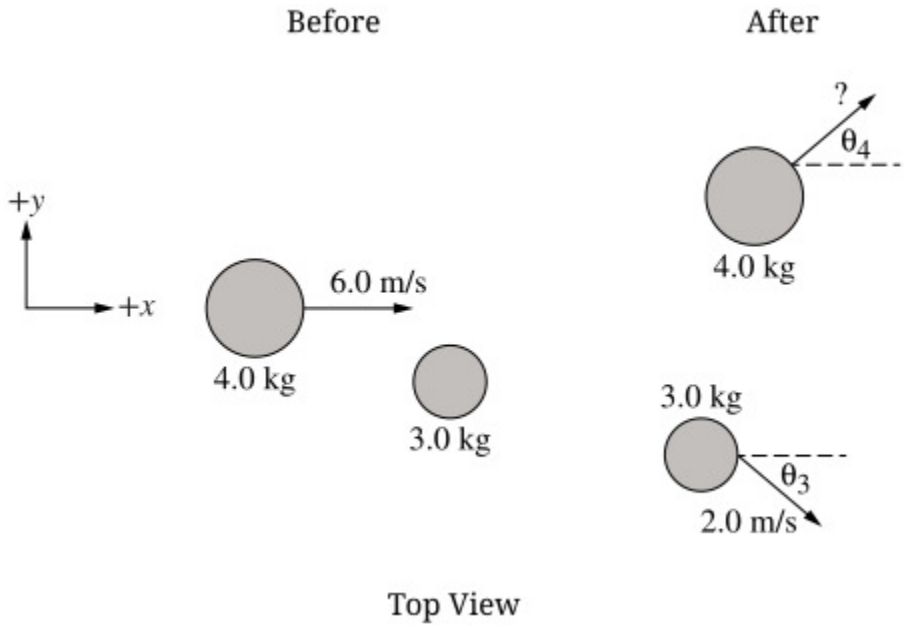
(B) $D + 4v(t - t_1)$

(C) $D + \frac{2}{3}v(t - t_1)$

(D) $D + \frac{4}{3}v(t - t_1)$

Unit 4 Topic Questions

28.



Two pucks are free to slide on a horizontal surface with negligible friction. The figure shows a top view of the pucks, one of mass $4.0\ \text{kg}$ and one of mass $3.0\ \text{kg}$, before and after they undergo an elastic collision with each other. Before the collision, the 4.0-kg puck slides at $6.0\ \text{m/s}$ in the $+x$-direction and the 3.0-kg puck is at rest. After the collision, the 3.0-kg puck moves with a speed of $2.0\ \text{m/s}$ at an unknown angle θ_3 measured clockwise from the $+x$-direction, as indicated, while the 4.0-kg puck moves at an unknown speed and at an unknown angle θ_4 measured counterclockwise from the $+x$-direction, as indicated. What is the speed of the 4.0-kg puck after the elastic collision?

(A) $1.5\ \text{m/s}$

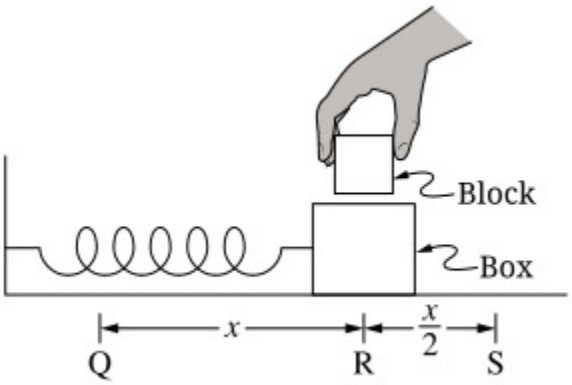
(B) $4.0\ \text{m/s}$

(C) $4.5\ \text{m/s}$

(D) $5.7\ \text{m/s}$

Unit 4 Topic Questions

29.



A spring has its left end fixed to a wall and its right end attached to a box, which can move back and forth on a horizontal surface with negligible friction, as shown in the figure. When the spring is at its relaxed length the box is at position R. The spring is compressed a distance x from its relaxed length and the box is then released from rest at position Q. The box moves from position Q to position R, continues moving on to position S a distance $\frac{x}{2}$ to the right of R, and continues moving beyond position S toward the right. As the box moves, a block can be dropped onto the box from a small height just above the box. When this happens, the block sticks to the box in a perfectly inelastic collision and the two objects move together horizontally after colliding. At which labeled position(s) can the block be dropped onto the box so that the block-box system's total momentum and total kinetic energy are each the same immediately before and immediately after the collision? Neglect the block's vertical component of motion, and treat the collision as occurring instantaneously.

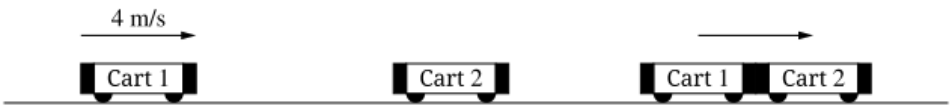
(A) At position Q only

(B) At position R only

(C) At position S only

(D) At positions Q, R, and S

30.



Two carts, Cart 1 of mass $0.25\ \mathrm{kg}$ and Cart 2 of mass $0.50\ \mathrm{kg}$, are free to move along a straight, horizontal track, as shown in the figure. There is negligible friction in the axles of the carts' wheels, and Cart 1 is initially moving to the right at $4\ \mathrm{m/s}$ before colliding with Cart 2, which is initially at rest. After the collision, the carts stick together and move to the right. What is the kinetic energy of the two-cart system immediately after the collision?

(A) $\frac{2}{3}\ \mathrm{J}$

(B) $1\ \mathrm{J}$

(C) $2\ \mathrm{J}$

(D) $6\ \mathrm{J}$

Unit 4 Topic Questions

31. Identical carts X and Y are placed on a straight, horizontal track, where they are free to move with negligible friction in the axles of their wheels. The carts undergo two separate collisions. In Collision 1, Cart X moves to the right with an initial kinetic energy K_i toward Cart Y, which is at rest, and the carts collide elastically. After the collision, Cart X comes to rest, and Cart Y moves to the right. In Collision 2, Cart X again moves to the right with the same initial kinetic energy K_i toward Cart Y, which is at rest, but the carts stick together during the collision and then move together to the right. Is the speed of Cart Y greater immediately after Collision 1 or immediately after Collision 2 ? What reasoning supports the claim?

(A) Immediately after Collision 1. During Collision 1, Cart Y gains an amount of kinetic energy equal to K_i, while in Collision 2, Cart Y gains an amount of kinetic energy less than K_i.

(B) Immediately after Collision 1. During Collision 1, Cart Y gains an amount of kinetic energy greater than K_i, while in Collision 2, Cart Y gains an amount of kinetic energy equal to K_i.

(C) Immediately after Collision 2. In Collision 1, the two-cart system loses more kinetic energy than in Collision 2.

(D) Immediately after Collision 2. When the carts stick together, Cart X transfers more energy to Cart Y than when the carts bounce off of one another.

32. Two blocks, one of mass m and one of mass $2m$, are free to move along a horizontal surface with negligible friction. The blocks undergo collisions in two separate scenarios. In the first scenario, the block of mass m moves with an initial speed v_0 toward the block of mass $2m$, which is initially at rest, and the blocks stick together after they collide. In the second scenario, the block of mass $2m$ moves with the same initial speed v_0 toward the block of mass m, which is initially at rest, and the blocks again stick together after colliding. The total kinetic energy of the two-block system decreases by an amount ΔK_1 in the first scenario and by ΔK_2 in the second scenario. What is the ratio $\frac{\Delta K_2}{\Delta K_1}$?

(A) $\frac{1}{3}$

(B) $\frac{1}{2}$

(C) $\frac{2}{3}$

(D) 1

33. Object 1 of mass m and Object 2 of mass $3m$ can slide along a straight, horizontal track with negligible friction. Object 1 initially moves to the right with speed v toward Object 2, which is initially at rest, and the two objects collide. Immediately after the collision, Object 1 moves to the left with speed $\frac{v}{2}$. Is this information sufficient to determine whether the collision was elastic? Why or why not?

(A) Yes, because the fact that the velocity of Object 1 changed direction during the collision means that the collision was not elastic.

(B) Yes, because conservation of momentum can be used to determine the final speed of Object 2, and the total kinetic energy of the two-object system after the collision can be calculated.

(C) No, because the final speed of Object 2 cannot be determined without knowing whether the collision was elastic or inelastic.

(D) No, because the collision can be elastic for many different values of the final speed of Object 2.

34. An 8-kg object is moving with a speed of $6\ \mathrm{m/s}$ along a horizontal surface while a 4-kg object is moving in the opposite direction with a speed of $3\ \mathrm{m/s}$ on the same surface. The objects collide and stick together after the collision. Friction between the objects and the surface is negligible. What is the decrease in total kinetic energy of the two-object system during the collision?

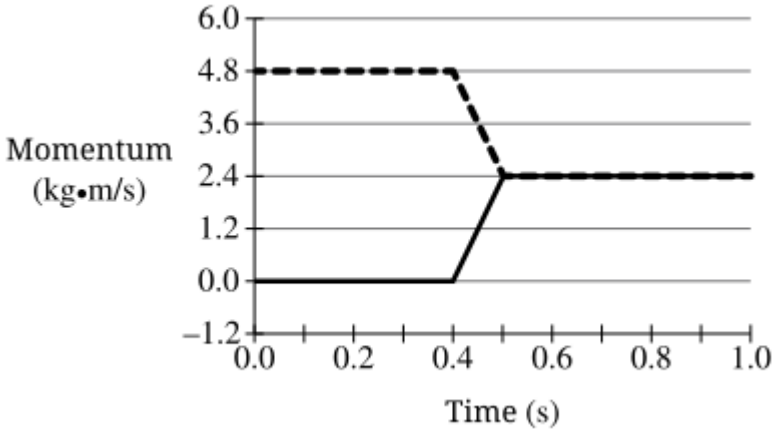
Unit 4 Topic Questions

(A) 54 J
(B) 90 J
(C) 108 J
(D) 162 J

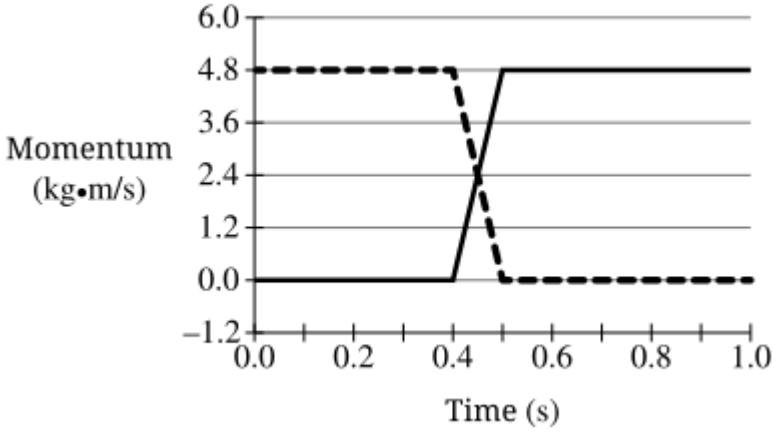
35. Two objects of equal mass undergo a series of collisions. For each collision, a graph is created of the momenta of both objects as functions of time. Which of the following graphs represents an elastic collision?

Unit 4 Topic Questions

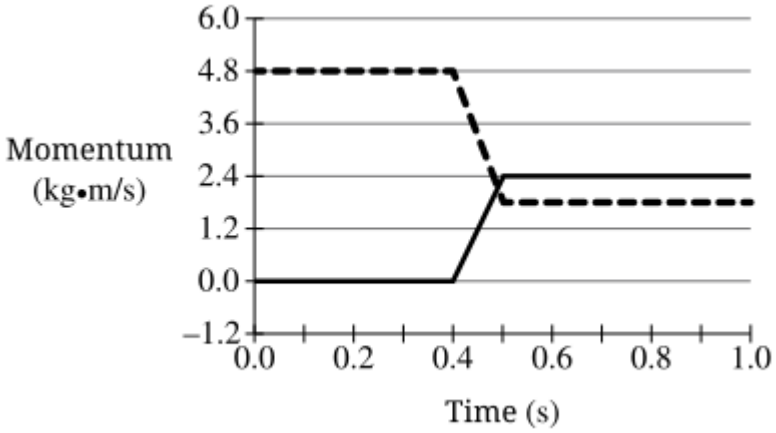
(A)



(B)

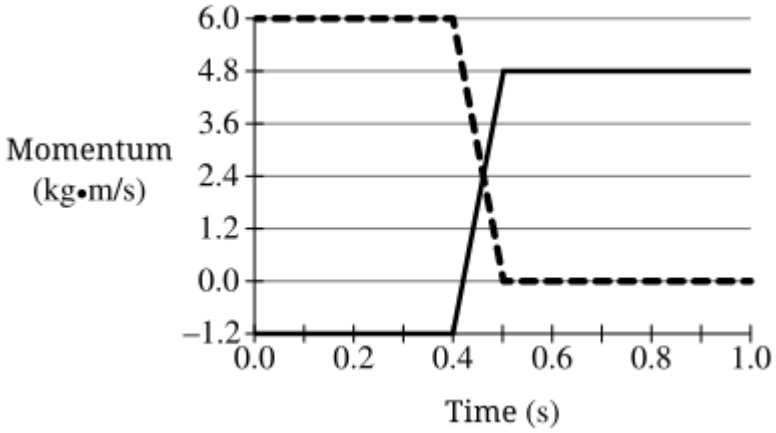


(C)



Unit 4 Topic Questions

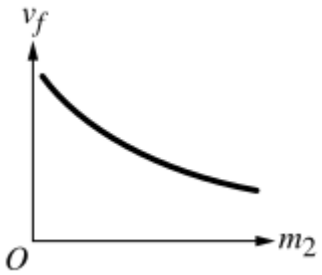
(D)



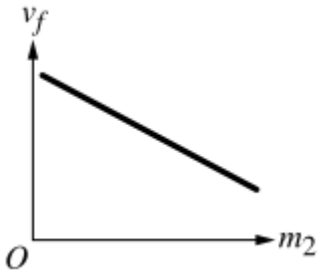
36. Students perform an experiment with multiple trials using blocks that can slide with negligible friction on a straight, horizontal track. In each trial, a block of mass m_1 slides with speed v_1 on the track and collides with a stationary block of mass m_2. The blocks stick together and move with speed v_f after the collision. In each trial, m_1 and v_1 are kept constant but the mass m_2 of the stationary block is varied, and v_f is recorded. The students graph v_f as a function of m_2. Which of the following could represent a best-fit curve to the data collected by the students?

Unit 4 Topic Questions

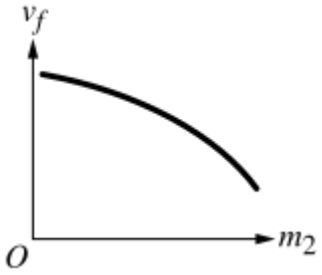
(A)



(B)



(C)



(D)

